SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I MS Naveen(192371002) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Naveen.ms

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

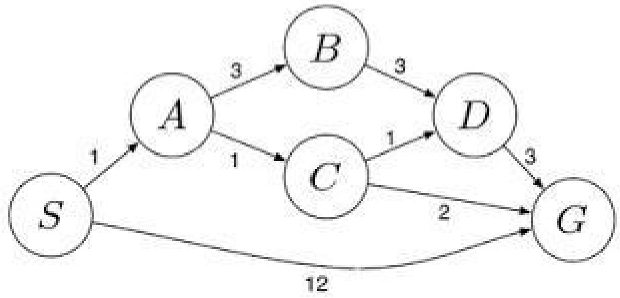
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Problem Understandin g	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
		requirements accurately			

1.Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

i. What are the percept's for this agent?

ii. Characterize the operating environment.

iii. What are the actions the agent can take?

iv. How can one evaluate the performance of the agent?

v. What sort of agent architecture do you think is most suitable for this agent? Why?

Mars Rover Agent – Solution

i. What are the percepts for this agent?

Percepts are the information the Mars rover receives from its sensors.

* Images from cameras
* Rock and soil composition data
* Temperature readings
* Atmospheric pressure
* Wind speed and direction
* Radiation levels
* Terrain and obstacle information
* Position and orientation (navigation sensors)
* Battery and power status

ii. Characterize the operating environment

The Mars rover operates in the following environment:

* Partially observable – It cannot observe the entire environment at once.
* Deterministic – Most actions have predictable outcomes, though terrain and weather may introduce uncertainty.

* Sequential – Current actions affect future actions.
* Dynamic – The environment can change due to dust storms, lighting, and terrain conditions.
* Continuous – Movement, time, and sensor values are continuous.
*Single-agent–The rover performs its tasks independently.

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iii. What are the actions the agent can take?

* Move forward, backward, left, or right
* Avoid obstacles
* Capture images and videos
* Collect rock and soil samples
* Drill into rocks
* Analyze samples using onboard instruments
* Send collected data to Earth
* Monitor battery and recharge using solar panels (if available)
* Stop or enter safe mode during hazardous condition.

iv. How can one evaluate the performance of the agent?

The rover's performance can be evaluated by:

* Accuracy of sample collection and analysis
* Successful completion of exploration tasks
* Distance covered safely
* Number of scientific discoveries
* Efficient energy (battery) usage
* Reliable communication with Earth
* Ability to avoid obstacles and hazards
* Mission duration without failure

v. What sort of agent architecture is most suitable? Why?

*Most suitable architecture:Utility-Based Agent (with learning capabilities)

*Reason:

* It evaluates different possible actions and selects the one that maximizes mission success.
* It balances multiple objectives such as safety, energy efficiency, and scientific value.
* It can adapt to uncertain and changing conditions on Mars.
* Learning capabilities help improve navigation and decision-making over time.

Conclusion: A **Utility-Based Learning Agent** is the best choice because it can make intelligent decisions in Mars' uncertain environment while maximizing scientific exploration and ensuring rover safety.

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- Problem-Solving Agent for OLA Cab Booking

- Type of Problem-Solving Agent

The most suitable agent is a **Goal-Based Agent**.

- Reason:

* The customer's goal is to **reach the destination**.

* The agent considers different cab options (Mini, Micro, Sedan, Prime, Shared, etc.) and selects the one that best satisfies the user's preferences such as cost, comfort, and availability.

□ Pseudocode

```text
START

Input pickup_location
Input destination
Input preferred_cab_type

Display available cabs

IF preferred_cab_type is available THEN
   Select preferred_cab_type
   Calculate fare
   Book the cab
   Display driver details and estimated arrival time
ELSE
   Display "Preferred cab is not available."
   Suggest other available cab types
   User selects another cab
   Calculate fare
   Book the cab
END IF

Display "Ride Confirmed"

STOP
```

- Working of the Agent

1. Customer enters the pickup and destination locations.
2. The agent searches for available cabs.
3. The customer selects a cab type (Mini, Micro, Sedan, Prime, Shared, etc.).
4. The agent checks availability.
5. If available, it books the cab and shows the fare and driver details.
6. If unavailable, it suggests other cab types.
7. The customer confirms the booking and the ride is scheduled.

Goal-Based Agent is the most suitable problem-solving agent because its primary objective is to help the customer reach the destination by selecting the most appropriate cab based on the user's preferences and availability.